Christopher Bussen

05/03/2024

Final Project Report

Dataset Selection

Similar to my midterm project, I wanted to study something related to sports so that I could combine my passion for data analysis with my love of sports. I have previously done projects on the MLB, NBA, and college basketball so I figured it would make the most sense to analyze some kind of NFL data for this project. I eventually found a dataset from Kaggle about NFL offenses from 2003-2023, which worked especially well as I wanted to be able to look at time series data for part of my project. This dataset involves statistics regarding team records, passing, rushing, penalties, turnovers, and more. Overall, the dataset contains 35 different variables across 672 observations, with each observation being a different team in a given season. The original author found all of the data on www.pro-football-reference.com and used a Python BeautifulSoup4 and a Selenium WebDriver to pull all of the HTML data and load it into a dataframe.

Data Preprocessing

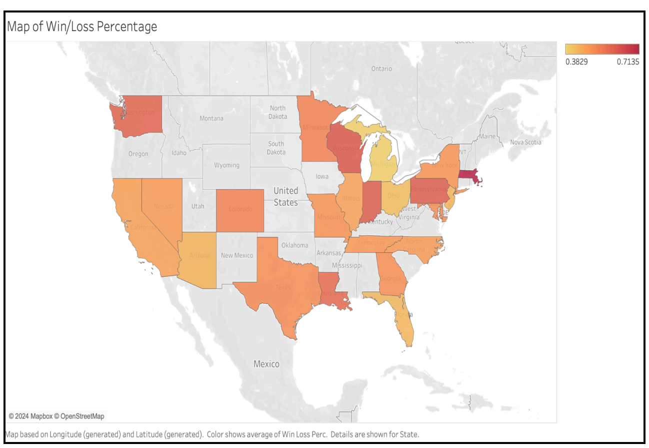
The process of cleaning and preprocessing my data took quite a while for multiple reasons. First off, I knew I wanted to add in location data for each team so that I would be able to create some sort of map. I originally tried to find a dataset that contained the location for each team that I would simply be able to merge with my dataset but was unsuccessful. My next thought was to find a webpage that contained all of this data in a table and create a web scraper to add it into my dataset. However, I already planned to use R to do more cleaning of this dataset so I instead opted to write a script in R to do so. This script simply looped through the dataset, checked the current team name, and then added the city and state where the team's stadium is located. This was done using a single for loop with nested if else statements. One thing I had to pay attention to here was that there were several teams that relocated during this time period, so I had to incorporate that into the script as well. Additionally, the (now) Washington Commanders were previously the Washington Football Team as well as the Washington Redskins. Due to the fact that they remained in the same location the entire time period, I decided to change all instances to the Washington Commanders for simplicity when working with the dataset.

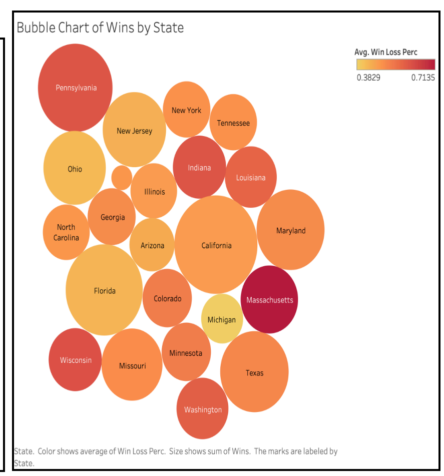
Another cleaning/preprocessing task I went through was adding stadium attendance to my dataset from this Kaggle dataset. This dataset provided weekly attendance numbers for each team from 2000-2019. Prior to merging this data into my original dataset, I realized that the dataset with attendance had a variable named “team” with the location part of the team name and a variable named “team_name” with the second half of the name (for example Arizona Cardinals would be split into Arizona and Cardinals). In order to properly merge the data, I had to change “team” to “tm" and then create a new variable named “team” with both parts of the team name pasted together (to match the variable in the original dataset for merging purposes). Once I did this, I created a variable for average attendance for a given team in each season by grouping the data by team and year and then using the mean() function. At this point I merged the average attendance variable into my original dataset by the “team” and “year” variables and wrote this to a new CSV file. It is worth noting that there were a good portion of observations with null weekly attendance due to the attendance dataset only going up until 2019. Despite this, I still thought it would be interesting to look at and I suspected that attendance for most teams has not drastically changed over the last four years so it was not crucial to attempt to find and add in this data.

Aside from the aforementioned tasks, I completed several more tasks to clean and preprocess the data. This included manually adding the attendance values for the Washington

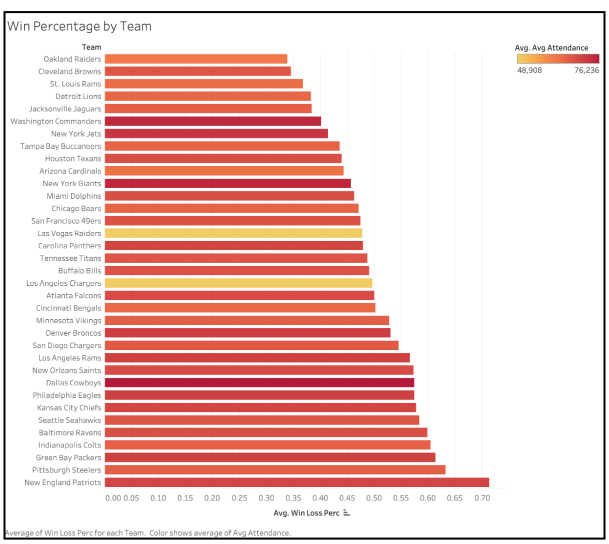
Commanders after realizing that the attendance dataset had them listed as the Redskins so it did not properly merge these values. I also realized that many of the values for the "ties" variable were null, so I replaced them with 0 (could have also just dropped the ties column with little to no impact). Lastly, upon trying to create a map for my visualizations, I realized that Tableau expected the data to be either a city or state but not a city followed by a state. As a result, I used Tableau Prep to split the location variable into one for city and another for state. It is also worth noting that, similar to the average attendance variable, the margin of victory variable also had a notable amount of null values. Despite attempts to find a good source to fill the rest of these values, I was unable to do so but decided to leave the variable as it could still provide interesting insights. Lastly, I originally planned to add team payroll information into my dataset. However, after spending several hours searching, I came to the conclusion that the NFL has not made this information readily available to the public, especially for seasons closer to the start of the 2000s. As a result, I had to move on without this variable.

<u>Exploratory Data Analysis</u>





To begin my exploratory data analysis, I decided to look at how the teams across different locations fared throughout the last 21 NFL seasons. In order to do so, I created a heatmap of the average win percentage by state. One of the first things that was important to note was which states had multiple teams and which states only had one. California has had the most teams since 2003 with five, followed by Florida with three and Texas, Missouri, Ohio, Pennsylvania, Maryland, and New Jersey each with two. This is noteworthy because many of these states had lower average win percentages due to one or more historically bad teams. For further transparency, I added the number of distinct (so that the same team from two different years does not count twice) teams and total combined wins to the tooltip when hovering over each state. The map shows that Massachusetts easily has the highest average win percentage just above 70%, with Wisconsin, Indiana, and Pennsylvania (only state among them with multiple teams) following at just above 60%. While I prefer to look at an actual map when digesting data related to location, I also decided to create a bubble chart displaying the same information for audience members who might prefer this visualization (this also works well because, while it does use location data, the actual location itself is not vital to visualization). In this bubble chart, I once again use color to represent the average win percentage while the size of each bubble represents the combined number of wins for teams throughout the state.



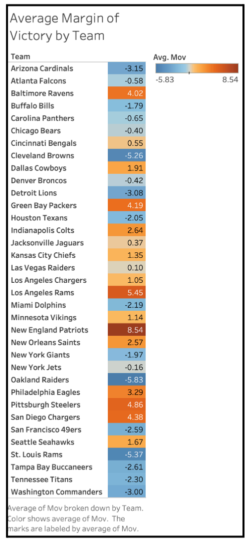
Christopher Bussen

05/03/2024

Next I decided to take a more in-depth look at each individual team's win percentage since 2003. In addition, I wanted to analyze how win percentage might be related to average attendance, so I added an attendance dimension using a color scale with the existing bar chart. At first glance, the bar chart shows that the Oakland Raiders, Cleveland Browns, St. Louis Rams, Detroit Lions, and Jacksonville Jaguars are the only teams with a win percentage below 40% since 2003. An interesting fact to point out here is that two of those teams (Raiders and Rams) have both been relocated recently. Similarly, the team with by far the lowest average attendance is the Los Angeles Chargers, who were relocated from San Diego to Los Angeles (which already had a team). Of the aforementioned bottom five teams, only the Browns, who are known for having a very loyal fanbase, have a solid average attendance number. On the other hand, the majority of teams in the top half of winning percentage generally have pretty good attendance, with the Cowboys, Broncos, and Packers leading the way. We can also see that the Commanders, Jets, and Giants all have very good average attendance numbers compared to other teams, but we must take into account the fact that their stadiums all have larger capacities than many other teams. All of this helps to show that while winning can impact attendance to an extent, there are many more factors to consider.

Christopher Bussen

05/03/2024

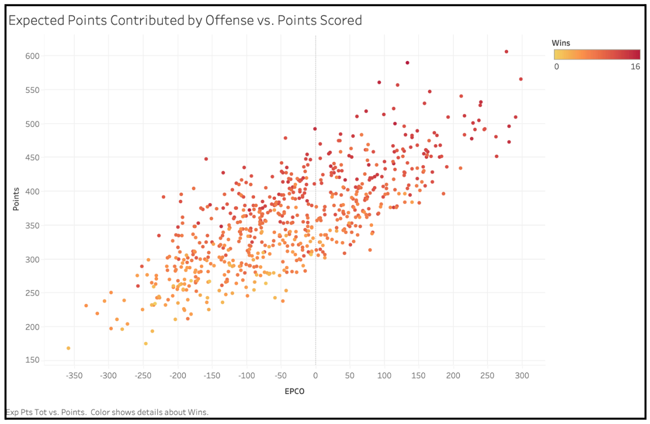


Average Margin of Victory by Team

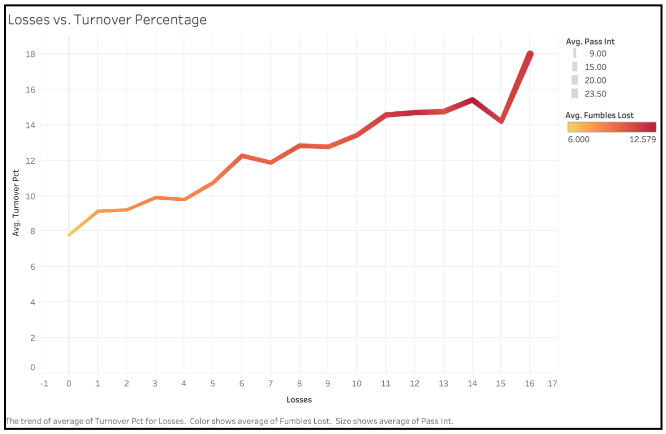
Team	
Arizona Cardinals	-3.15
Atlanta Falcons	-0.58
Baltimore Ravens	4.02
Buffalo Bills	-1.79
Carolina Panthers	-0.65
Chicago Bears	-0.40
Cincinnati Bengals	0.55
Cleveland Browns	-5.26
Dallas Cowboys	1.91
Denver Broncos	-0.42
Detroit Lions	-3.08
Green Bay Packers	4.19
Houston Texans	-2.05
Indianapolis Colts	2.64
Jacksonville Jaguars	0.37
Kansas City Chiefs	1.35
Las Vegas Raiders	0.10
Los Angeles Chargers	1.05
Los Angeles Rams	5.45
Miami Dolphins	-2.19
Minnesota Vikings	1.14
New England Patriots	8.54
New Orleans Saints	2.57
New York Giants	-1.97
New York Jets	-0.16
Oakland Raiders	-5.83
Philadelphia Eagles	3.29
Pittsburgh Steelers	4.86
San Diego Chargers	4.38
San Francisco 49ers	-2.59
Seattle Seahawks	1.67
St. Louis Rams	-5.37
Tampa Bay Buccaneers	-2.61
Tennessee Titans	-2.30
Washington Commanders	-3.00

Average of Mov broken down by Team. Color shows average of Mov. The marks are labeled by average of Mov.

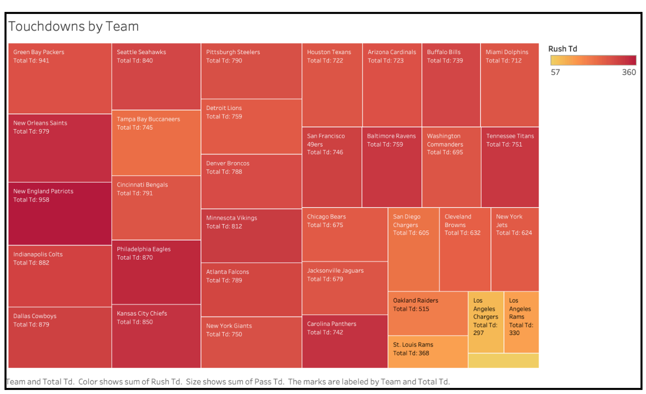
While I had already looked at which teams were winning most often in the given time span, I thought it would be interesting to look at how easily teams were winning (or losing) games. Therefore I decided to analyze each team's average margin of victory over the last 21 seasons. For this visualization, I thought it would be best to keep things simple as I only wanted to look at one measure, so I created a table. I decided to add in a diverging color scheme to make it easier for an audience to spot how teams differ and especially to be able to tell the positive and negative margins of victory apart. From this table, it becomes clear that the Patriots are typically beating teams by the most points at around 8.5 points per game. They are followed by the LA Rams at about 5.5, the Steelers around 4.9, and the San Diego Chargers around 4.4. The Rams and Chargers being this high despite not being in the top 10 for win percentage likely means that they have generally had high quality offenses that allow them to win by a lot of points but their defense can also consistently give up a lot of points as well. On the other end of the spectrum, we see the Oakland Raiders with an average margin of victory of -5.8, the St. Louis Rams at -5.4, and the Browns at -5.3. This is not shocking seeing as these are the bottom three when looking at average win percentage as well.



Moving onto the next visualization, I wanted to look at expected points contributed by the offense (EPCO) as it was a statistic I was previously unfamiliar with. After a little bit of research, I discovered that this statistic basically represents how well an offense performs compared to what is expected. More specifically, it looks at the expected number of points for a drive depending on the situation and compares that value to the number of points the drive actually ends with. For example, if a drive starts on the 50 yard line, the offense might be expected to come away with three points. If they end up scoring a touchdown, this would mean the offense contributed four points (7-3=4). This explanation can be further broken down to play-by-play scenarios as well. I decided to plot EPCO against the number of points a team actually scored in a season, assuming there would be a somewhat noticeable linear correlation as performing better than expected on each play/drive (higher EPCO) should in turn result in more points. The created scatter plot displays this pattern quite clearly, as increasing EPCO appears to lead to an increase in points. Furthermore, I decided to see how this might impact the number of wins a team has so I added a color scheme for wins. This additional dimension displays how, for the most part, a smaller EPCO leads to less wins and vice versa, which makes sense as an over-performing offense would have a better chance of winning games. With that being said, there is also a large amount of variation in win totals among the teams in the middle of the pack due to varying defensive quality mattering more for average offenses.



While wins are very important to look at when analyzing different NFL teams, I also wanted to look into some statistics people tend to focus less on that have a negative impact on a team's performance. Thus, the next visualization displays the relationship between losses and turnover percentage through a line chart. More specifically, it shows the average team turnover percentage for each possible number of losses in a season and a partial makeup of the turnover percentage. Average passing interceptions is shown by the line thickness whereas average fumbles lost is shown by the line color. Not surprisingly, we see that as a team has more losses, we typically see a higher turnover rate, with the potential to almost double. Similarly, both interceptions and fumbles lost will drastically increase on average for a team with more losses.

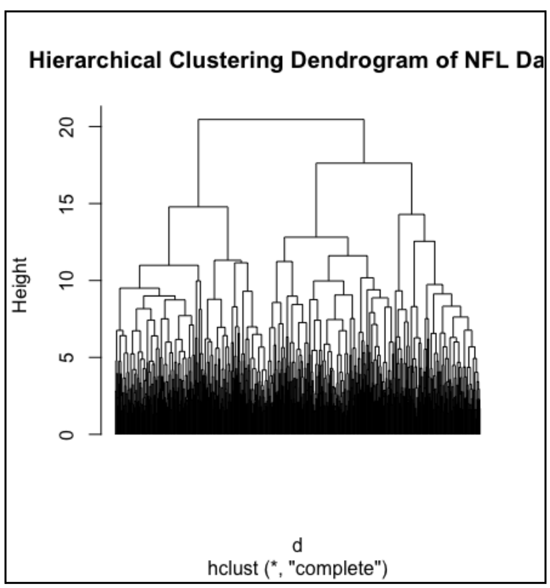


The last visualization I created (other than my time-series plots) breaks down which teams have been scoring the most touchdowns over the last two decades using a tree map. Each section of the tree map is labeled with a team and its total number of touchdowns. Additionally, the size of each section is based on passing touchdowns while the color is based on rushing touchdowns. Right away, we see that the top three teams in passing touchdowns are the Packers, Saints, and Patriots. This makes sense as they have each had hall of fame quarterbacks until recently (Aaron Rodgers, Drew Brees, and Tom Brady respectively). It is also easy to see that the Patriots have the most rushing touchdowns and are followed by the Eagles and Saints. Lastly, the bottom six in total touchdowns are the three teams that have been relocated recently and their counterparts which is expected.

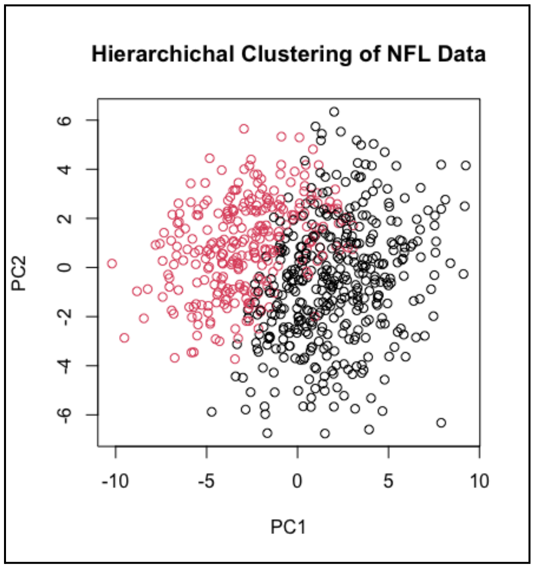
Dimensionality Reduction and Clustering

After performing exploratory data analysis and discovering all of the insights and conclusions mentioned above, I utilized R and applied PCA in order to reduce the dimensionality of my dataset. This was especially useful because my dataset is high dimensional with 38 different variables after my additions, which can make it difficult to analyze as a whole. Prior to performing PCA, I had to consider the restraints I needed to impose on my dataset. Knowing that PCA cannot handle categorical data, I had the option to either encode team name, city, and state or drop the columns. I ultimately decided to drop them because I anticipated they would not have a major impact on the components (also encoding would have been complex with that many values). Next, I had to check which variables had null values and decide how to handle them as PCA cannot be applied with null values. While it was not ideal, I ended up removing the margin of victory and average attendance columns because it would have been extremely time-consuming to find the information and fill in all of the null values. Lastly, after handling categorical and null values, I decided to scale the data because many of the variables operated on different scales. At this point, I performed PCA on the remaining data and analyzed the summary. Based on the output, I decided to use the first nine principal components as they capture about 90% of the variance. I also wanted to analyze what variables were most important in constructing these principal components, so I looked at the rotation matrix for the PCA results, which tells you how much each variable contributes to each principal component. For each of principal components one through nine, the top contributors were points, pass attempts, year, turnovers, penalty yards committed, yards per play on offense, games, first downs by penalty, and fumbles lost respectively.

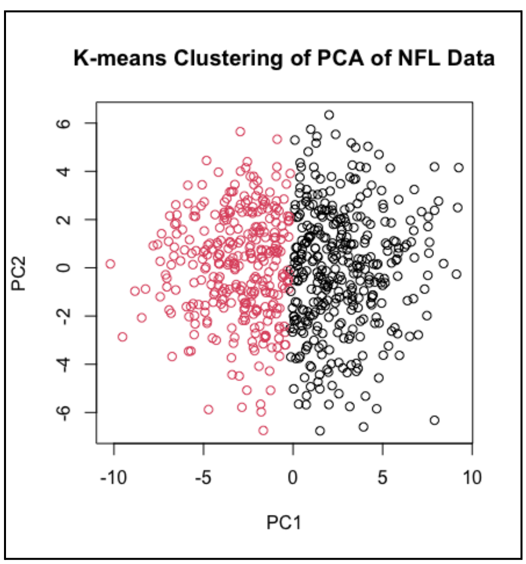
Once I had my PCA results ready, I decided to start off with hierarchical clustering because I was not sure how many clusters would be appropriate for the first nine principal components. After calculating the distance matrix for the principal components and performing hierarchical clustering, I created the following dendrogram:



From this dendrogram, it appeared that either two or three clusters would work best, so I tried both and two clusters seemed to give better results, yielding the following plot:



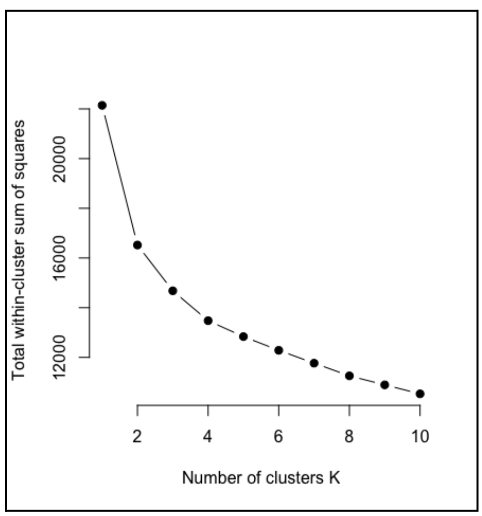
Next, I decided to perform K-means Clustering to see how it would compare to hierarchical clustering. Based on the previous results, I decided to define two centers in the kmeans() function and then plotted the results to get the following:



Christopher Bussen

05/03/2024

It was immediately clear that K-means Clustering does a better job at separating the data than hierarchical clustering. Based on these clustering results using the PCA data, I interpreted the clusters as grouping NFL offenses into above average and below average categories. This is because all of the principal components are made up of different offensive statistics, so these would be the two logical groupings. I originally thought it might group offenses into three groups (bad, average, and good) so I tried K-means Clustering with three centers and it provided worse results as there was much less distinction between clusters and more overlapping points. This helped me categorize different teams/observations in the dataset throughout the EDA process as it shifted my idea of the tiers of teams. Finally, I wanted to double check that I was using the correct number of clusters so I analyzed Silhouette Scores and utilized the elbow method. Three clusters gave an average Silhouette Score of 0.19 whereas the score for two clusters was higher at 0.24. For the elbow method, I created the following plot:

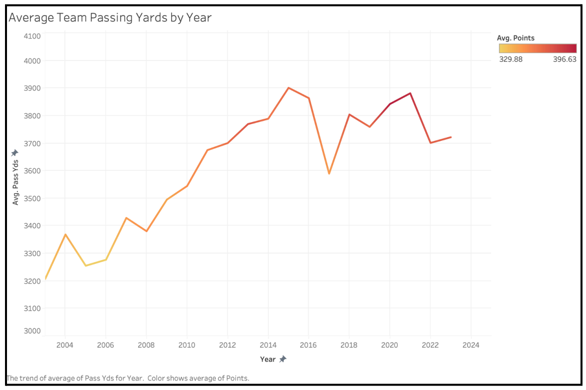


While the elbow does appear to be closer to three clusters than two, I did not think it was a significant enough drop to warrant switching to three clusters based on all of the other results I had that suggested two clusters would perform better.

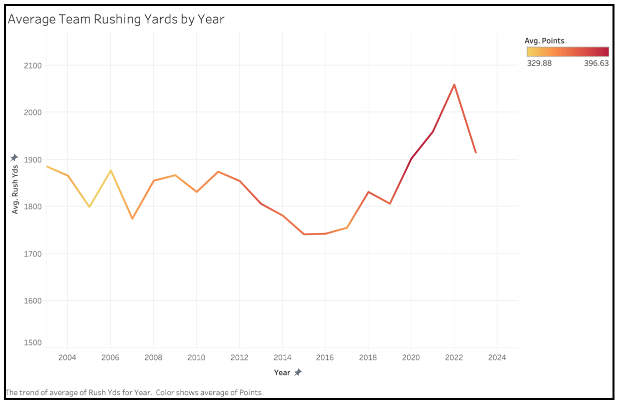
Christopher Bussen

05/03/2024

Time Series Analysis



As a result of my data involving different statistics from all NFL teams over a 21 year span, I was able to utilize time series data to analyze offensive trends throughout the league. I was primarily interested in looking at the progression of both passing and rushing yards, which are two of the most basic offensive statistics in football. I thought this would be intriguing because many people are adamant that the NFL has intentionally given many advantages to offenses over the last 20 years because a good offense generates more fan interest. I began by looking at the relationship between time and average team passing yards throughout a season using a line chart. To add more context, I decided to show average points in a season using color. While it was hard to see the trends at first glance, changing the scale on the y-axis drastically helped. From there, it became clear that, while the change is not drastic, both average passing yards and average points per team in a season have been consistently trending up. In 2020, average passing yards was almost 650 yards higher than the 2003 season and average points was over 60 higher (around 40 yards and 4 points more per game). Moreover, with the NFL recently increasing the season from 16 games to 17, these numbers will likely continue to increase. In terms of seasonality, there is not much to discuss as these stats come from separate seasons and are averaged over all teams (might be easier to look at if there were month to month stats due to weather during games). When looking at anomalies, we do see one rather unexpected drop in 2017 by almost 300 passing yards. This can likely be attributed to an unexpected increase in quarterback injuries, leading to more backup quarterbacks playing.



After analyzing the trends in passing yards, I did the same for rushing yards. However, contrary to the conclusions for passing yards, average rushing yards per year has not experienced any notable changes. While there are definitely fluctuations from year to year (as expected), average rushing yards has mostly stayed between 1800 and 1900 yards per team each year, with lows dipping to 1740 rushing yards in 2015. So, overall it appears that average rushing yards are remaining mostly constant rather than trending up. Similar to passing yards, there is not much analyzing to do when it comes to seasonality. When discussing anomalies, we can see spikes in both 2021 and 2022, but these are somewhat expected due to the NFL adding another game to the season as mentioned above. Finally, while I did not look into this because I wanted to investigate league-wide trends rather than individual teams, the year-by-year values for teams are likely to have some autocorrelation due to the same player often playing quarterback/running back for multiple years in a row.

<u>Technical Documentation</u>

Christopher Bussen

05/03/2024

Throughout this process, I used Tableau to create all of my visualizations outside of my plots during clustering. As mentioned before, I used a mix of R/RStudio, Tableau Prep, and Excel to clean and preprocess the data. Tableau Prep allowed me to easily split my location field into city and state while Excel allowed me to easily modify small numbers of specific observations. R was especially useful for filling in the location fields of my data as well as merging the attendance data into my dataset. In order to fill location fields in R, I created a for loop with nested if else statements to check the team name. To merge attendance into my dataset, I created a team variable that matched the other dataset, calculated average attendance for each time in each season, then merged the variable in by team and year. I will attach screenshots of code snippets for both of those processes below:

```
for (i in 1:nrow(df)) {
   curr <- df[i,]
   if (curr$team == "Arizona Cardinals"){
      df[i,]$location <- "Glendale, Arizona"
   } else if (curr$team == "Atlanta Falcons"){
      df[i,]$location <- "Atlanta, Georgia"
   } else if (curr$team == "Baltimore Ravens"){
      df[i,]$location <- "Baltimore, Maryland"
   } else if (curr$team == "Buffalo Bills"){
      df[i,]$location <- "Orchard Park, New York"
   } else if (curr$team == "Carolina Panthers"){
      df[i,]$location <- "Charlotte, North Carolina"
   } else if (curr$team == "Chicago Bears"){
      df[i,]$location <- "Chicago, Illinois"
   } else if (curr$team == "Cincinnati Bengals"){
      df[i,]$location <- "Cincinnati, Ohio"
   } else if (curr$team == "Cleveland Browns"){
      df[i,]$location <- "Cleveland, Ohio"
   } else if (curr$team == "Dallas Cowboys"){
      df[i,]$location <- "Arlington, Texas"
```

```
attendance <- attendance %>%
   rename(
      tm = team
   )

attendance$team <- paste(attendance$tm, attendance$team_name)
```

Christopher Bussen

05/03/2024

```r
avg_attendance <- attendance %>%
   group_by(team, year) %>%
   summarize(avg_attendance = mean(weekly_attendance, na.rm = TRUE))

merged_w_attendance <- left_join(df, avg_attendance, by = c("team", "year"))

write.csv(merged_w_attendance, "nfl.csv", row.names = TRUE)
```

I also used R to perform dimensionality reduction using PCA via the prcomp() function within the stats library. Furthermore, I used the hclust() and kmeans() functions from the stats library in R to perform clustering techniques on the data after applying PCA to reduce dimensionality. It is important to understand that PCA cannot handle categorical variables, hence they were excluded. Additionally, it cannot handle null values so variables with large amounts of nulls were dropped prior to PCA. I will display screenshots of this code below as well - for explanations of key steps, please see the comments above most of the lines of code:

```r
# don't use team and location - categorical - drop rows w NA
data <- select(data, -c(team, city, state))

# check which columns have NA
colSums(is.na(data))

# drop columns w NAs - mov and Avg attendance
data <- data[, colSums(is.na(data)) == 0]

# scale data due to different scales
data_scaled <- scale(data)

# perform pca on scaled data
pca_results <- prcomp(data_scaled, center = TRUE, scale. = TRUE)
summary(pca_results)
# View loadings (rotation matrix)
(loadings <- pca_results$rotation)

# plot first two dimensions
plot(pca_results$x[,1:2], xlab = "PC1", ylab = "PC2", main = "PCA of NFL Data")

# 9 PCs capture 90% of variance
pcs <- pca_results$x[, 1:9]
```

```r
# perform hierarchical clustering
d <- dist(pcs)
hc <- hclust(d, method = "complete")

# plot dendrogram
plot(hc, labels = FALSE, hang = -1, main = "Hierarchical Clustering Dendrogram of NFL Data")

# cut tree and plot clusters
clusterCut <- cutree(hc, 2)
plot(pca_results$x[,1:2], col = clusterCut,
     xlab = "PC1", ylab = "PC2", main = "Hierarchichal Clustering of NFL Data")

# perform kmeans
set.seed(2024)
(kmeans_result <- kmeans(pcs, centers = 2))

# plot clusters
plot(pca_results$x[,1:2], col = kmeans_result$cluster,
     xlab = "PC1", ylab = "PC2", main = "K-means Clustering of PCA of NFL Data")
```

Christopher Bussen

05/03/2024

Lastly, I used both Silhouette Scores and the elbow method (both methods of evaluating the quality of clustering results) to help decide how many clusters I should use in K-means Clustering. To do so, I used both the silhouette() function as well as the kmeans() function once again. I will display the code snippet for these below:

```r
# methods to decide clusters
# silhouette scores to see how well data is clustered
silhouette_scores <- silhouette(kmeans_result$cluster, d)
plot(silhouette_scores, col = 1:3, border = NA)


# elbow method
wss <- sapply(1:10, function(k) {
   kmeans(data_scaled, centers = k, nstart = 25)$tot.withinss
})

# plot elbow
k_values <- 1:10  # Range of k values used
plot(k_values, wss, type = "b", pch = 19, frame = FALSE,
     xlab = "Number of clusters K", ylab = "Total within-cluster sum of squares")
```

The full code for both data cleaning/preprocessing as well as applying PCA and clustering techniques can be found within the submitted zip file.